$\frac{Kg \times V}{J \cdot s}$

$\frac{1}{cm^3} \times C \times \frac{cm^2}{s}$ $= \frac{C}{cm^2 s}$ A/cm²

$V_d = \mu E =$

$I = neV_dA$

$J = neV_d \Rightarrow J =$

$J = \sigma E$

School of Engineering and Engineering Technology
Department of Information Communication Engineering
First Semester Test 2023/2024 Session
EEE 301/303 Electronic Engineering I

**Unit 2.0**

**Instruction:** Answer **all Questions**

**Time: 90 minutes.**

$\frac{cm^2}{V \cdot s} \times \frac{cm^2}{s} = \frac{cm^4}{Vs^2}$

**Question One**

(a) Explain the concept of band structure and doping in electronic engineering. **(2 marks)**

(b) Using a sketch diagram, explain the significance of the band gap as related to the conductivity of insulator, semiconductor, and conductor materials. **(3 marks)**

(c) Assume electronic charge $q = 1.6 \times 10^{-19}$ C, $KT/q = 20$mV, and electron mobility $\mu_n = 1000 cm^2/V\text{-}s$ if the concentration of electrons injected into a p-type silicon sample is $2 \times 10^{21}/cm^4$ .

(i) Compute the magnitude of electron diffusion current density in $A/cm^2$ given that $k = 1.38 \times 10^{-23}$J **(5 marks)**

$\frac{KT}{q}\mu_n \Rightarrow D_n$ $20 \times 10^{-3} \cancel{V} \times \frac{10^3 cm^2}{\cancel{V}s}$

$\left.\frac{20 \, cm^2}{s}\right\} E$

**Question Two**

(a) Explain the significance of Fermi energy in electronic engineering.

(b) In the Fermi-Dirac distribution function, state the value of f (E) using a simple equation in the following cases and explain what each parameter represents:

(i) At T = 0K (zero Kelvin) where E > EF, **(2marks)**

(ii) When E = EF. **(2marks)**

(c) Determine the ρ (E) and the approximate number of quantum states (n) using the density of states (DOS) for electrons with energy between 4.0 -5.0 eV in a metal with volume 1.4 cm³. Given: $m = 9.11 \times 10^{-31}$kg, $1eV = 1.6 \times 10^{-19}$J, $h = 6.626 \times 10^{-34}$ J-s **(4 marks)**

(d) From the value obtained in (c), compute the:

(i) Fermi wave ~~vector~~ Velocity $V_f$ **(2 marks)**

**Question Three**

(a) In the Ebers-Moll model, derive an expression for $I_E$ and $I_B$. Define each parameter in the derived equations. **(2 marks)**

(i) For a certain D-MOSFET, $I_{DSS}$= 10mA, and $V_{GS}$(off) = -8V. Determine whether this is an n-channel or p-channel MOSFET.
(ii) Compute $I_D$ at $V_{GS}$ = -4V. **(2 marks)**

(iii) Calculate $I_D$ at $V_{GS}$ = 4V. **(1 mark)**

(iv) Sketch the characteristic curve using all obtained values. **(1mark)**

(b) State and explain three characteristics of logic families. **(2 marks)**

(c) Draw a simple truth table for a TTL NAND circuit for cases 1 and 2. **(2 marks)**

School of Engineering and Engineering Technology
Department of Information Communication Engineering
First Semester Examination 2023/2024 Session
EEE 301 Electronic Engineering I — Unit 2.0

**Instruction:** Answer only **Four (4) Questions** — **Time: 150 minutes.**

$\sigma = qE(\mu_n)$

## Question One

(a) As a CPE student enrolled in the course EEE 301 Electronic Engineering, clarify the ambiguities of student X from the MNE department regarding the conductivity of different materials by following these steps:

$I = NeV_dA$
$J_n = NeV_d$
$V_d = \mu E$

(i) Explain the concepts of drift and diffusion current as they relate to semiconductor materials. **(4 marks)**

$\sigma E$
$\sigma = q(\mu_n + \mu_p)$

(ii) Use a sketch diagram to explain the significance of the band gap in relation to the conductivity of insulator, semiconductor, and conductor materials. **(4 marks)**

$qE\mu_n$

(iii) Explain the concept of band structure and doping in electronic engineering. **(3 marks)**

$E = 1eV$

(b) A DC voltage of 10V is applied across an n-type silicon bar having a rectangular cross- section and a length of 1cm. The donor doping concentration $N_D$ and the mobility of electrons $\mu_n$ are $10^{16}$ $cm^3$ and $1000cm^2/V\text{-}s$ , respectively.

$V_d = \frac{dist}{time}$

(i) Determine the electric field (E) for the semiconductor. **(3 marks)**

(ii) Calculate the average time (in $\mu s$) taken for electrons to move from one end of the bar to the other. **(5 marks)**

(c)Given electronic charge $q = 1.6 \times 10^{-19}$ C, $KT/q = 25mV$, and electron mobility $\mu_n = 1000cm^2/V\text{-}s$, if the concentration of electrons injected into a p-type silicon sample is $1 \times 10^{21}/cm^4$ .

(i) Compute the magnitude of electron diffusion current density in $A/cm^2$ given that $k = 1.38 \times 10^{-23}J$ **(6marks)**

## Question Two

(a) In Eber-Moll model, derive an expression for $I_E$, $I_C$ and $I_B$ and define each parameter in the derived equations. Include a sketch diagram to aid in the explanation. **(12marks)**

(b) Consider a silicon sample with a unit cross-sectional area with a total length $X = 1\mu m$, which is in thermal equilibrium. Given the following information: T= 300K, electron charge = $1.6 \times 10^{-19}$ C, $N_D = 10^{16}$ $cm^3$, thermal voltage $(KT/q) = 26mV$, and electron mobility = $1350cm^2/V\text{-}s$

(i) Determine the magnitude of electric field at $X = 0.5\mu m$ **(4marks)**

(ii) Calculate the magnitude of the electron drift density at $X = 0.5\mu m$ **(4marks)**

(iii) Provide a reason for the final answers obtained in (i) and (ii). **(2marks)**

(iv) Explain how to determine the total current for N-type and P-type semiconductors in relation to (i) and (ii) using simple equations, and describe what each parameter represents in each of the stated equations. **(3 marks)**

**Question Three**

(a) Define the density of states (DOS) and elaborate on it significance in the context of semiconductor materials **(3marks)**

(b) In the Femi-Dirac distribution function, state the value of f (E) using simple equation in the following scenarios, and explain the significance of each parameter:

(i) At T = O K (zero Kelvin) where E > EF, **(3marks)**

(ii) At T = O K (zero Kelvin) where E < EF and **(3marks)**

(iii) When E = EF **(2marks)**

(c) Determine the ρ (E) and the approximate number of quantum states (n) using density of state (DOS) for electrons with energies between 5.0 and 6.0 eV in a metal with a volume of $1.2cm^2$

Given: $m = 9.11x10^{-31}kg$, $1eV = 1.6x10^{-19}J$, $h = 6.626 \times 10^{-34}$ J-s **(6marks)**

(d) Using the values obtained in (c), calculate the following:

(i) Femi wave vector, $K_f$ $\frac{J^2}{S^2} \times \frac{1}{Cm^2} \times \frac{1}{Kg}$ **(2marks)**

(ii) Femi energy, $E_f$ **(2marks)**

(iii) Femi momentum $P_f$ and mxv $\frac{J}{S}$ **(2marks)**

(iv) Femi velocity, $V_f$ Kgm/s **(2marks)**

**Question Four**

(a) A 200-level chemistry student with some ambiguities wants to test your understanding of semiconductor materials. As a CPE student enrolled in EEE 303, explain the following to the student:

(i) Explain the difference between the intrinsic and extrinsic semiconductors **(2marks)**

(ii) State the mass action law in semiconductors and represent it in a simple equation. Explain what each parameter means in the equation **(4marks)**

(b) Find the conductivity of silicon, in intrinsic condition at room temperature (300K), given that ni for Si is $1.5 \times 10^{10}/cm^3$, $\mu_n = 1300cm^2/V\text{-}s$ and , $\mu_p = 500cm^2/V\text{-}s$ **(6marks)**

(c) Determine the concentration of holes and electrons in a p-type Si at 300K, if the conductivity is 300S/cm. Given ni = $1.5\times 10^{10}/cm3$ and $\mu_p = 500cm^2/V\text{-}s$. Provide your commentary on the values obtained in (c) here **(7marks)**

(d) Explain briefly, with the aid of a sketch diagram, how a transistor is formed. Give examples of major transistors. **(6marks)**

**Question Five**

(a) A student seeks clarifications on MOSFET. In your explanation, address the following:

(i) Explain the circuit operation of a D- MOSFET with the aid of diagram. **(6marks)**

(ii) Sketch a well-labelled transfer characteristics curve of a D- MOSFET. **(3marks)**

(iii) List the modes in which the Ebers-Moll model could be utilized **(2marks)**

(b) Consider a D- MOSFET, $I_{DSS}$ =10mA, and $V_{GS}$ (off) = -8V

(i) Determine whether this is an n-channel or p-channel MOSFET **(2marks)**

(ii) Compute $I_D$ at $V_{GS}$ = -3V **(3marks)**

(iii) Calculate $I_D$ at $V_{GS}$ = 3V **(3marks)**

(iv) Sketch the characteristic curve using all values obtained and provide commentary on this curve **(6marks)**

**Question Six**

(a) The data sheet for E- MOSFET provides $I_D$ (on) = 500mA at $V_{GS}$ = 10V and $V_{GS}$ (th) = 1V.

Determine:

$\sigma_p = q\mu_p n.$

(i) The resulting value of K and **(2marks)**

(ii) The drain current for $V_{GS}$ = 5V **(3marks)**

(iii) Plot the transconductance curve for the device **(3marks)**

(b) Provide two examples for each of the following logic families:

(i)BJT saturated logic family (ii) BJT unsaturated logic family (iii) Unipolar logic family **(3marks)**

(c) State and explain six characteristic of logic families **(6marks)**

(d) Illustrate the standard circuit for a TTL NAND gate, explain its operation and identify two characteristics that make TTL preferable to DTL or RTL **(8marks)**